Day 2: Photosynthesis Practice Questions

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Q1 Where in the chloroplast do the light-dependent reactions occur, and which wavelengths of light are mainly absorbed by chlorophyll?

Q2 What molecule is split during photolysis, and what by-product is released?

Q3 What is the role of the enzyme RuBisCO in the Calvin cycle?

Q4 State the overall chemical equation for photosynthesis and list the three limiting factors.

Answers

They occur in the thylakoid membranes. Chlorophyll absorbs mainly blue and red light.

Water (H2O) is split during photolysis, releasing oxygen (O2) as a by-product.

In the stroma, RuBisCO fixes carbon dioxide onto RuBP.

Equation: 6 CO2 + 6 H2O + light energy -> C6H12O6 + 6 O2. Limiting factors: light intensity, carbon dioxide concentration, and temperature.